Chapter 0 Notes

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1 Quantum Fields

Particles and waves in quantum field theory are described as excitations of a quantum field. A field can be thought of as a mathematical object that takes a spacetime position as input and spits out an "amplitude" as an output. Physically, fields are like gravitational or electromagnetic fields permeating throughout spacetime. So an excitation of a field is energy being put into the field, placing it in a higher energy state much like how an electron in hydrogen can be excited from the ground state. The field is neither a particle nor a wave solely, but a duality of the two, an idea that will become apparent in chapter 2.

2 Lorentz Transformations

Since we have mentioned spacetime as an input for a quantum field, it is an inescapable feature that spacetime means special relativity. Quantum field theory must work in the framework of special relativity in that all physics looks the same even if we shift inertial reference frames. Hence, Lorentz transformations, ones that shift from one inertial frame to another, are naturally considered:

$$\bar{t} = \gamma\Big(t - \frac{vx}{c^2}\Big)$$

$$\bar{x} = \gamma(x - vt)$$

$$\bar{y} = y$$

$$\bar{z} = z$$

A derivation of these transformations can be found at `https://www.physics.umd.edu/~yakovenk/teaching/Lorentz.pdf`, courtesy of Victor M. Yakovenko.

In matrix form, the Lorentz transformations can be cast into

$$\begin{pmatrix} c\bar{t} \\ \bar{x} \\ \bar{y} \\ \bar{z} \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

where $\gamma = 1/\sqrt{1-\beta^2}$ and $\beta = v/c$ and in index notation, into $\bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu$. x^ν is a four-vector and transforms appropriately under $\Lambda^\mu{}_\nu$. In fact, one can generalize $\Lambda^\mu{}_\nu$ as $\frac{\partial \bar{x}^\mu}{\partial x^\nu}$ such that any tensor obeys the tensor transformation law

$$\bar{T}^{i'\dots k'}_{l'\dots n'} = \frac{\partial \bar{x}^{i'}}{\partial x^i} \cdots \frac{\partial \bar{x}^{k'}}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^{l'}} \cdots \frac{\partial x^n}{\partial \bar{x}^{n'}} T^{i\dots k}_{l\dots n}$$

For the sake of clarity, all that is needed is the matching of indices. Lower indices cancel out upper ones and vice versa. The partial derivatives cancel out certain indices and create new ones, transforming each index in $T^{i\dots k}_{l\dots n}$ entirely by a full set of coordinate transformations.

Now, some other examples of four-vectors (going back to four-vectors) are the energy-momentum four-vector $p_\mu = \langle E, \vec{p}\rangle$ and the vector four-potential $A_\mu = \langle \phi, \vec{A}\rangle$. We can take the inner product of p_μ with itself facilitated by the metric convention $g^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$.

$$p_\mu p^\mu = E^2 - \vec{p}^2 = m^2$$

This the energy-momentum relation with $c = 1$. Notice the μ indices cancel out to highlight to us that this is a Lorentz scalar, meaning the relation holds true in any reference frame.

3 Maxwell's Equations

A_μ can reformulate electromagnetism in tensor notation. By setting $\vec{B} = \nabla \times \vec{A}$ and $\vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}$, the equations may read

$$\partial_\alpha F^{\beta\alpha} = -\mu_0 J^\beta$$

for the laws

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{and} \quad \nabla \times \vec{B} = \mu_0 \vec{J} + \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}$$

and

$$\partial_{[\alpha} F_{\beta\gamma]} = 0$$

for the laws

$$\nabla \cdot \vec{B} = 0 \quad \text{and} \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

where the electromagnetic tensor in the metric we are working with is

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & -B_x & 0 \end{pmatrix}$$

and that $J_\beta = \langle c\rho, \vec{J} \rangle$ is the four-current. Also, $\partial_\mu = \langle \frac{1}{c}\partial_t, \vec{\nabla} \rangle$.

4 Fourier Transforms

The one-dimensional Fourier transform is given by

$$\tilde{f}(\omega) = \int dt\; e^{i\omega t} f(t).$$

3B1B has provided a stunning geometrical explanation that illustrates what exactly a Fourier transform does in the context of sound frequencies. Find it at `https://www.youtube.com/watch?v=spUNpyF58BY`. Essentially, shifting ω allows $e^{i\omega t}$ to act like a measuring stick on $f(t)$. If there is overlap between the phase factor $e^{i\omega t}$ and the function $f(t)$ for a particular ω, the integral would produce a larger value. The phase factor effectively sifts out the strong "frequencies" that exist in $f(t)$.

Its four-dimensional spacetime extension is simply

$$\tilde{f}(\omega, \vec{k}) = \int d^4x\; e^{i(\omega t - \vec{k}\cdot\vec{x})} f(t, \vec{x}).$$

For QFT, Fourier transforms are handy in simplifying the mathematics. They are tools that ease differential equations into algebraic ones. They will be necessary.

Practice Questions

1. Velocity Addition Formula

Frame O' moves at a velocity v' relative to the stationary observer O. Frame O'' moves at a velocity v'' relative to O'. The Lorentz transformation of another Lorentz transformation is still a Lorentz transformation. Using this property of Lorentz transformations, see if you can find what the velocity of O'' is in respect to observer O.

2. Bianchi Identity

Explicitly, the Bianchi identity is written as

$$\begin{aligned}\partial_{[\alpha}F_{\beta\gamma]} &= \frac{1}{3!}(\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} - \partial_\alpha F_{\gamma\beta} - \partial_\beta F_{\alpha\gamma} - \partial_\gamma F_{\beta\alpha}) \\ &= 0.\end{aligned}$$

Simplify this to

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

Then, for specific values of α, β, and γ, show that Faraday's Law, $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$, can be extracted.

3. The Dirac Delta

Compute the Fourier transform of the two-sided decaying exponential $f(t) = e^{-a|t|}$. Once you find the Fourier transform $\tilde{f}(\omega)$, normalize it to a total area of 1. The limit of $a \to 0$ for $\tilde{f}(\omega)$ gives the Dirac delta distribution $\delta(\omega)$. Thus, prove that

$$\delta(\omega) = \frac{1}{2\pi}\int dt\; e^{i\omega t}.$$